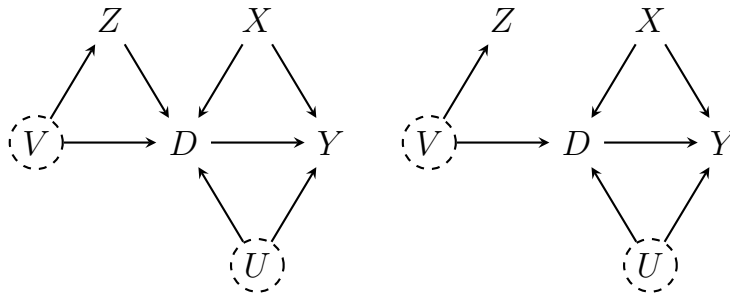


Controls for IV Estimation

Validity of Instrumental Variable in the Special Case

In Assignment 3, we have a special scenario where there is an unobserved V that affects D and Z (left panel). Can Z be a valid instrument? Here I simulate the DGP and check the property of the estimate. Note the true β in the DGP is 5.



```
library(AER)
library(ggplot2)
library(dplyr)
library(tidyr)
set.seed(1)

## run simulation for 200 times
results <-
  data.frame(iteration=1:200,
             OLS=rep(NA,200),
             IV=rep(NA,200))
for (i in 1:200){

  ## define unobserved
  V <- rnorm(1000, mean=3, sd=1)
  U <- rnorm(1000, mean=-2, sd=3)

  ## define observed
  Z <- 1 * V + runif(1000, min=0, max=2) + rnorm(1000, mean=0, sd=2)
  X <- rnorm(1000, mean=5, sd=3) + rnorm(1000, mean=0, sd=1)
```

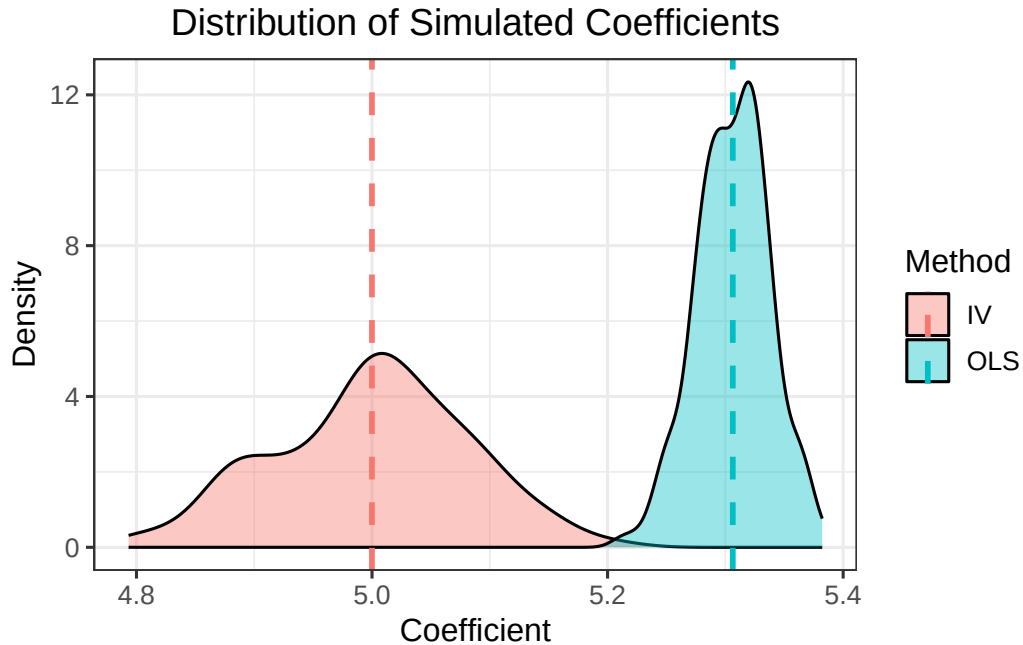
```

D <- -5 * V + 2 * Z + 0.5 * U + 1 * X + rnorm(1000, mean=0, sd=1)
Y <- 5 * D + 1 * X + 2 * U + rnorm(1000, mean=0, sd=1)

## save results
results[i,"OLS"] <- summary(lm(Y ~ D + X))$coef[2,1]
results[i,"IV"] <- summary(ivreg(Y ~ X + D | X + Z))$coef[3,1]
}

## plot distribution
results %>%
  pivot_longer(c(OLS, IV), names_to = "Method", values_to = "Coefficient") %>%
  group_by(Method) %>%
  mutate(mean_coef = mean(Coefficient)) %>%
  ggplot(aes(x = Coefficient, fill = Method)) +
  geom_density(alpha = 0.4) +
  geom_vline(aes(xintercept = mean_coef, color = Method),
             linetype = "dashed", size = 1) +
  labs(title = "Distribution of Simulated Coefficients",
       x = "Coefficient", y = "Density") +
  theme_bw(base_size = 12) +
  theme(
    plot.title = element_text(hjust = 0.5),
    axis.text.x = element_text()
  )

```



The simulation results show that you don't have to control for V (it is unobserved in any case) or X (X does not affect Z) to use Z as a valid IV, as long as V does not affect Y .

Now what about another scenario where the relationship between Z and D is completely driven by unobserved V , as in the right panel? Yes you still can!

```
set.seed(1)

## run simulation for 200 times
results <-
  data.frame(iteration=1:200,
             OLS=rep(NA,200),
             IV=rep(NA,200))
for (i in 1:200){

  ## define unobserved
  V <- rnorm(1000, mean=3, sd=1)
  U <- rnorm(1000, mean=-2, sd=3)

  ## define observed
  Z <- 1 * V + runif(1000, min=0, max=2) + rnorm(1000, mean=0, sd=2)
  X <- rnorm(1000, mean=5, sd=3) + rnorm(1000, mean=0, sd=1)
```

```

D <- -5 * V + 0.5 * U + 1 * X + rnorm(1000, mean=0, sd=1)
Y <- 5 * D + 1 * X + 2 * U + rnorm(1000, mean=0, sd=1)

## save results
results[i,"OLS"] <- summary(lm(Y ~ D + X))$coef[2,1]
results[i,"IV"] <- summary(ivreg(Y ~ X + D | X + Z))$coef[3,1]
}

## plot distribution
results %>%
  pivot_longer(c(OLS, IV), names_to = "Method", values_to = "Coefficient") %>%
  group_by(Method) %>%
  mutate(mean_coef = mean(Coefficient)) %>%
  ggplot(aes(x = Coefficient, fill = Method)) +
  geom_density(alpha = 0.4) +
  geom_vline(aes(xintercept = mean_coef, color = Method),
             linetype = "dashed", size = 1) +
  labs(title = "Distribution of Simulated Coefficients",
       x = "Coefficient", y = "Density") +
  theme_bw(base_size = 12) +
  theme(
    plot.title = element_text(hjust = 0.5),
    axis.text.x = element_text()
  )

```

